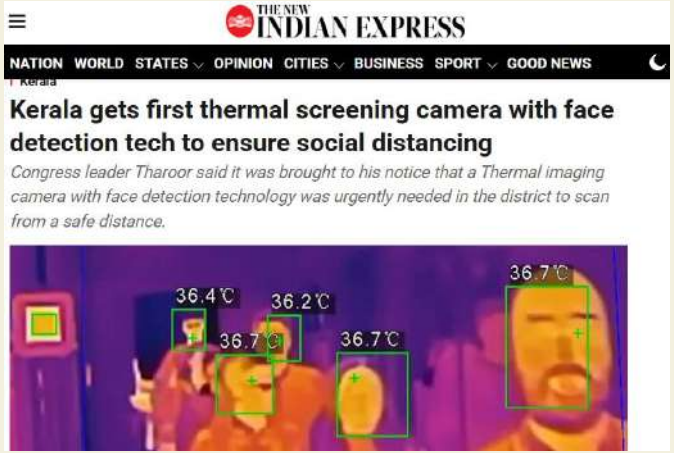
Potential Harms Caused by AI Systems

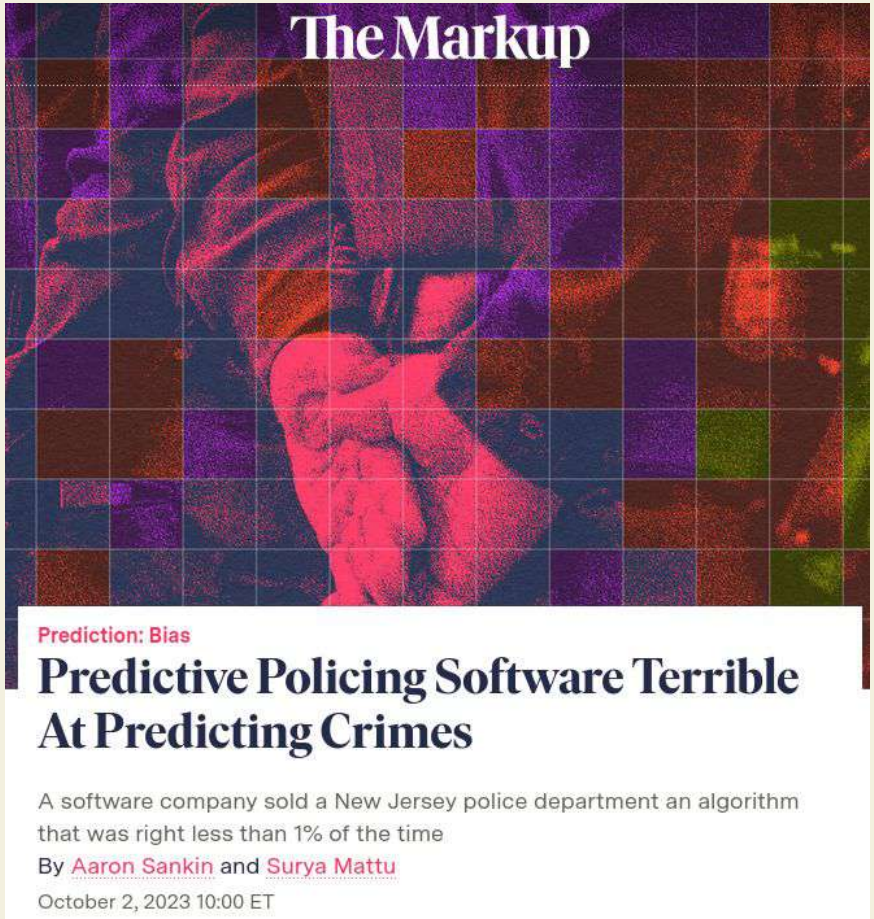
Facial recognition Technology in India



- AadhaarFaceRD app: Pensioners scan face monthly to confirm "alive" status.
- Poshan Tracker (National Nutrition Mission - Anganwadi/ICDS).
- Examination Centers: Preventing impersonation during competitive exams.
- Karnataka Assembly Elections (2023): Pilot FRT requiring voters to install app for facial verification.

FRT and Predictive Policing and Surveillance

what’s wrong with policing by algorithm?



Predictive policing

Predictive policing uses computer systems to analyze large sets of data to help decide where to deploy police or to identify individuals who are purportedly more likely to commit or be a victim of a crime.

- the use of arrest data to predict geographical crime hotspots.
- mined social media data that can be used to determine how likely it is that someone might commit a crime.

The Hyderabad Police has created the Integrated People Information Hub, a multifaceted '360 degree' profile of all residents, to 'predict' people to be arrested, based on 24/7 surveillance.

The Hyderabad model of CCTV surveillance

Telangana is leading the way in using tech to track citizens. Other Indian cities are looking to emulate the model
Several cities are already in the process of setting up integrated command-and-control centres. The new hardware, combined with AI and facial recognition, could easily be misused

Varsha Bansal
Updated • 10 Nov 2020, 07:34 AM IST

Credit Card

Police pitch this AI-based integrated safe city solution as a necessary means for social control and maintaining order, but many don't buy it.(Photo:)

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Hyderabad: The Perils of Importing 'Hi-Tech Policing' Without the Necessary Safeguards

Srinivas Kodali
18/Mar/2023 • 5 min read

The lack of accountability mechanisms along with the proliferation of surveillance technologies is causing various problems, which neither the police nor the political establishment wants to address.

Representative image of Hyderabad police. Photo: PT

Telangana Is Inching Closer to Becoming a Total Surveillance State

The CCTV network, paired with facial recognition technology, an ever-expanding database of biometric information and a willing police force, has the potential to monitor its citizens to a worrying degree.

- Delhi Police's Crime Mapping Analytics and Predictive System (CMAPS), which determines crime 'hotspots' in the city.
- to decide how much force is required and where it needs to be employed.

Hindustan Times

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Preventing crime before it happens: How data is helping Delhi Police

Hindustan Times | By Karn Pratap Singh, New Delhi

Feb 27, 2017 06:38 PM IST

The software being used by Delhi police accesses real-time data from 100 helpline and, using ISRO's satellite imageries, spatially locates the calls and visualises them as cluster maps to identify crime hot spots.

Data in New Delhi's Predictive Policing System

Vidushi Marda*
vidushi@article19.org
Article 19

Shivangi Narayan*
shivangi.narayan@gmail.com
Jawaharlal Nehru University

ABSTRACT

In 2015, Delhi Police announced plans for predictive policing. The Crime Mapping, Analytics and Predictive System (CMAPS) would be implemented in India's capital, for live spatial hotspot mapping of crime, criminal behavior patterns and suspect analysis. Four years later, there is little known about the effect of CMAPS due to the lack of public accountability mechanisms and large exceptions for law enforcement under India's Right to Information Act. Through an ethnographic study of Delhi Police's data collection practices, and analysing the institutional and legal reality within which CMAPS will function, this paper presents one of the first accounts of smart policing in India. Through our findings and discussion we show what kinds of biases are present within Delhi Police's data collection practices currently and how they translate and transfer into initiatives like CMAPS. We further discuss what the biases in CMAPS can teach us about future public sector deployment of socio-technical systems in India and other global South geographies. We also offer methodological considerations for studying AI deployments in non-western contexts. We conclude with a set of recommendations for civil society and social justice actors to consider when engaging with opaque systems implemented in the public sector.

CCS CONCEPTS

• **Social and professional topics** → *Governmental regulations; Race and ethnicity; Cultural characteristics;* • **Computing methodologies** → *Reasoning about belief and knowledge.*

KEYWORDS

Fairness-Aware Machine Learning, Predictive Policing, Interdisciplinary, Sociotechnical systems

ACM Reference Format:
Vidushi Marda and Shivangi Narayan. 2020. Data in New Delhi's Predictive Policing System. In *FAT* '20: Proceedings of ACM Conference on Fairness, Accountability, and Transparency, January 27–30, 2020, Barcelona, Spain.* ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/3351095.3372865

1 INTRODUCTION

Award for its use of facial recognition in crime solving [30]. The National Crime Records Bureau of India recently published a tender for the Automated Face Recognition System which would be used for "criminal identification, verification and its dissemination among various police organizations and units across the country" [16]. One of the first initiatives towards the use of machine learning (ML) in law enforcement was pioneered by the Delhi Police. For context, Delhi is a city with a population of close to 30 million people. It is the national capital, and is also anecdotally referred to as the 'rape capital' of India due to historically high records of violence against women. This is important to understand to follow the larger safety discourse in Delhi and why policing interventions of a certain kind may gain public legitimacy. In 2015, Delhi police announced the use of the Crime Mapping, Analysis and Mapping System (CMAPS), a predictive policing system [20] that would access data directly from the Dial 100 call centre to plot the geographic location of calls and calculate crime hotspots.

The limitations and dangers of predictive policing have been studied in jurisdictions like the United States and Europe. It is well understood that these systems risk over-policing vulnerable populations [15] and exacerbate problematic institutional biases [14]. The implementation and quotidian experience of these systems in the global South (or post-colonial) jurisdictions like India remain to be studied. It is particularly crucial to study such deployments in India at this juncture because of the ongoing efforts at wholesale deployment of various ML technologies in all aspects of public

Use of Delhi Police's Crime Mapping Analytics and Predictive System (CMAPS)

Also, in response to an RTI filed in 2022, the Delhi Police acknowledged that facial recognition technology was used to investigate "over 750 cases related to the North East Delhi riots" and that the results were presented as evidence against those arrested. That accounts for at least 98.9% of all riot-related cases (758 in total) being "solved" with the help of facial recognition technology.

Business Standard In 2019

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Delhi police facial recognition software has only 2 per cent accuracy: HC told

How were they 'identified'?

The investigating officer in the Ratan Lal Murder Case, Inspector Gurmeet Singh of the Crime Branch, told *The Wire* that the case was "solved" using advanced technologies, including video and image enhancement tools (Amped FIVE by Amped Software) and facial recognition software (AI Vision by Innefu Labs). He confirmed that all the accused named in the case were identified using these tools.

Five years since deadly Delhi riots, many police cases are falling apart

28 February 2025

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Umang Poddar

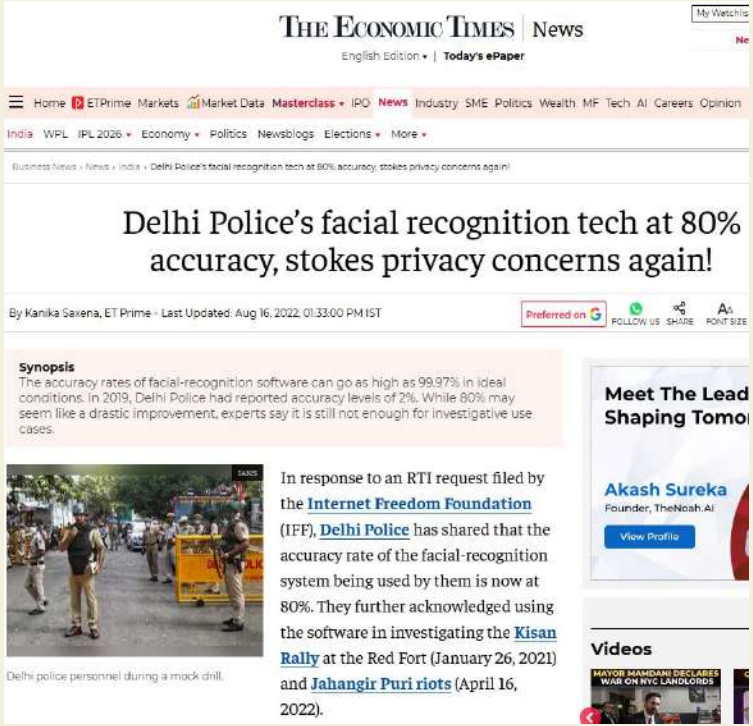
BBC Hindi, Delhi

https://www.bbc.com/news/articles/c757zl67n7wo

More than 80% of the cases related to the violence where courts gave decisions have resulted in not guilty or discharges.

https://pulitzercenter.org/stories/ai-took-over-policing-delhi-who-bore-brunt

70 arrests sound impressive—but what if 20% were wrong?



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Business News › News › India › Delhi Police's facial recognition tech at 80% accuracy, stokes privacy concerns again!

Delhi Police's facial recognition tech at 80% accuracy, stokes privacy concerns again!

By Kanika Saxena, ET Prime · Last Updated: Aug 16, 2022, 01:33:00 PM IST

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Synopsis

The accuracy rates of facial-recognition software can go as high as 99.97% in ideal conditions. In 2019, Delhi Police had reported accuracy levels of 2%. While 80% may seem like a drastic improvement, experts say it is still not enough for investigative use cases.

Delhi police personnel during a mock drill.

In response to an RTI request filed by the Internet Freedom Foundation (IFF), Delhi Police has shared that the accuracy rate of the facial-recognition system being used by them is now at 80%. They further acknowledged using the software in investigating the Kisan Rally at the Red Fort (January 26, 2021) and Jahangir Puri riots (April 16, 2022).

Meet The Lead
Shaping Tomo

Akash Sureka
Founder, TheNoah.AI
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MAYOR MAMDANI DECLARES WAR ON NYC LANDLORDS

https://economictimes.indiatimes.com/news/india/delhi-polices-facial-recognition-tech-at-80-accuracy-stokes-privacy-concerns-again/articleshow/93587861.cms?from=mdr

" 80% match is a similarity match.

The percentage means that the photo that is being matched with the photo in the database is 80% similar. Here, treating 80% similar as a positive identification means that we are treating 80% as 100%, which means that there is a 20% margin of error. "

- an expert from Internet Freedom Foundation (IFF)

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Punjab Police won smart policing award for Punjab Artificial Intelligence System

By Mohd Ujaley — On Jun 4, 2018 EGOV WATCH

INTERNET FREEDOM FOUNDATION

FRT | Madras HC | Surveillance | Facial Recognition

Madras HC issues notice on petition challenging use of FRT in Tamil Nadu

Madras High Court has issued notice in a petition challenging the installation and deployment of Facial Recognition Technology ("FRT") across the State.

Gayatri Malhotra

21 August, 2023
6 min read

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Smart policing: Nailing criminals not a handicap anymore

PK Jaiswar Tribune News Service Amritsar, September 9 Even as the city police have initiated digital mapping to bring down crime rate in the holy city, the police have failed to use the Punjab Artificial Intelligence System (PAIS) at its...

TRIBUNE WEB DESK
Updated At : 12:56 PM Sep 10, 2021 IST

Madras HC issues notice in a legal challenge to Facial Recognition Technology

Facial Recognition

The petition argues that the FRT deployment by the TN Govt. infringes the right to privacy, is not backed by law, and fails the proportionality standard laid by the SC.

Privacy concerns

ALJAZEERA

News | Cybersecurity

India targeted high-profile journalists with Pegasus spyware: Amnesty

Investigation shows journalists in India face 'threat of unlawful surveillance' along with other 'tools of repression'.

Pegasus can be used to access a phone's messages and emails, peruse photos and eavesdrop on calls [File: Dado Ruvic/Illustration/Reuters]

Discrimination

MIT Technology Review

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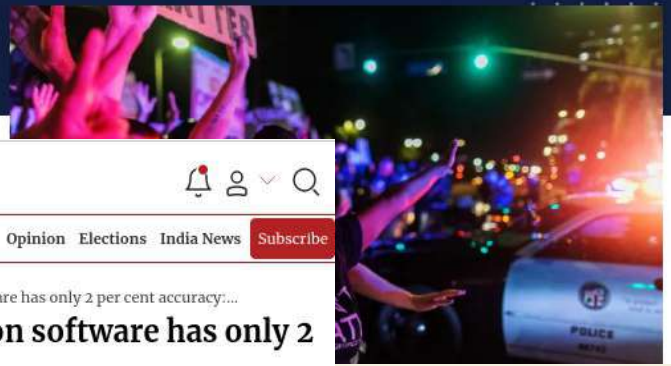
POLICY

Predictive policing is still racist —whatever data it uses

Training algorithms on crime reports from victims rather than arrest data is said to make predictive tools less biased. It doesn't look like it does.

By Will Douglas Heaven

February 5, 2021



Lack of accuracy

Business Standard

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GEOGRAPHIES IN DEPTH

India is using facial-recognition to reunite missing children with their families

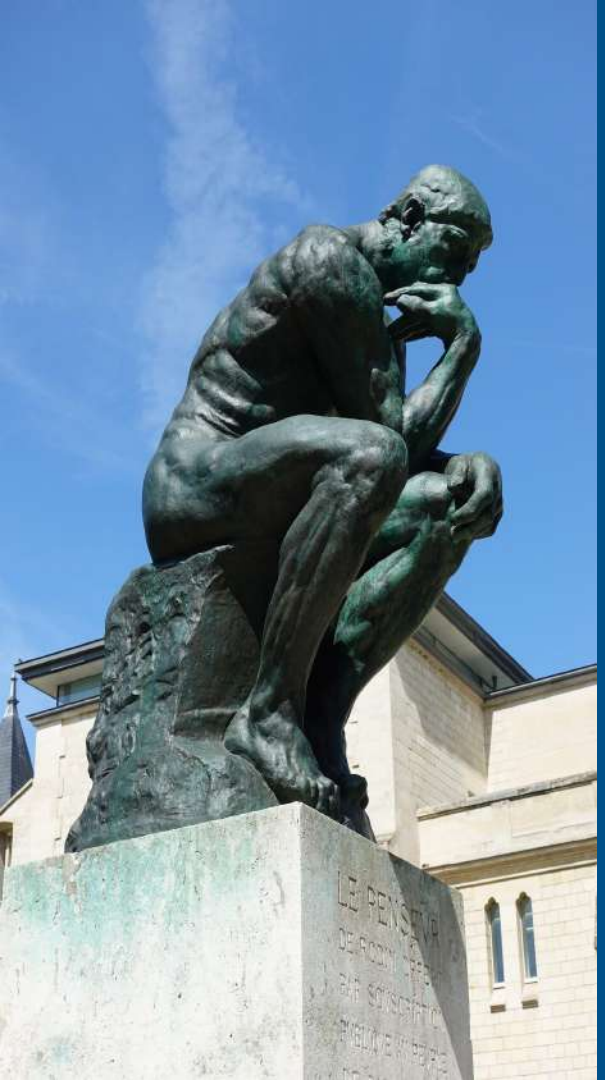
May 11, 2018

https://www.weforum.org/stories/2018/05/india-facial-recognition-technology-find-missing-children-lion/

Police in New Delhi trialled facial recognition technology and identified almost 3,000 missing children in four days.

The software scanned the pictures of 45,000 children living in children's homes and orphanages.

Between April 6 and 10 2018, it was able to match 2,930 children with photographs held on the lost child database run by the government.



Should we use Facial Recognition Technology everywhere at Academic institutes like IITPKD?

If used, what are the advantages and ethical concerns that may arise?

How can we use AI ethically to make our cities safer from crime?